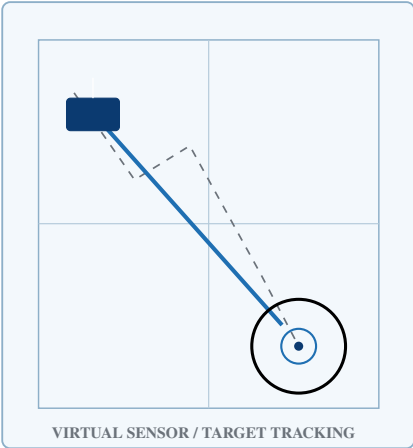


PROJECT

HORIZON

AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals



VIRTUAL SENSOR / TARGET TRACKING

DEVELOPED BY	TEAM LEAD	TEAM ID	PROBLEM ID
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PROBLEM STATEMENT TITLE

Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals

DOCUMENT DESIGNATION

USER MANUAL

Research Grade Edition

Installation | Operation | Configuration | Monitoring

PROJECT	TEAM	TEAM ID	PROBLEM ID
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SYSTEM REQUIREMENTS & INSTALLATION

1 System Requirements

HORIZON is supplied as a Python-based desktop application with a PySide6 graphical interface and command-line execution support. The following environment is taken from the current HORIZON release configuration.

1.0.1 Software Requirements

Table 1: Software requirements for the current HORIZON release.

Component	Technology	Version	Operational Purpose
Programming Language	Python	3.11+	Application execution
Numerical Computing	NumPy	≥ 1.26 , < 2.0	Numerical operations and random-number handling
Computer Vision	OpenCV	≥ 4.9 , < 5.0	Image-processing operations
Graphical Interface	PySide6 (Qt6)	≥ 6.6 , < 7.0	HORIZON operator console
Configuration	PyYAML	≥ 6.0 , < 7.0	YAML configuration parsing
Neural Inference	ONNX Runtime	≥ 1.16	Beacon-detection inference
Visualization	Matplotlib	≥ 3.8 , < 4.0	Graph and report generation
PDF Generation	ReportLab	≥ 3.6 , < 4.0	Automated report generation
Testing	pytest	≥ 8.0 , < 9.0	Software verification tests

User Note Install the dependencies from the supplied `requirements.txt` file. The version ranges above describe the software environment specified for the current HORIZON release.

1.0.2 Hardware Requirements

Table 2: Minimum and recommended execution environment.

Specification	Minimum	Recommended
CPU	Quad-core, 2.5 GHz	8-core, 3.5 GHz
RAM	8 GB	16 GB
GPU	Not required	—
Storage	2 GB free	5 GB free
Display	1920×1080	2560×1440
Operating System	Windows 10, 64-bit	Windows 11

1.1 Project Directory

The HORIZON project is organized into separate application, configuration, simulation, control, PAT, estimation, benchmark, and test areas.

```
coarse-align-x/
|-- main.py                # CLI + GUI entry point
|-- configs/default.yaml  # Validated baseline YAML
|-- requirements.txt       # Pinned dependencies
|-- simulator/
|   |-- core/              # Clock, config, seed, engine
|   |-- world/             # 2000x2000 world, beacon
|   |-- trajectories/      # 6 trajectory models
|   |-- camera/            # Viewport, camera model
|   |-- disturbances/      # 4-stage disturbance pipeline
|   |-- perception/        # Classical + Neural + Hybrid
|   |-- tracking/          # IMM-EKF, track management
|   |-- scenarios/         # 13 scenario definitions
|   +-- ui/                # 5-screen PySide6 GUI
|-- control/               # Gimbal controllers
|   |-- pid.py
|   |-- adrc_controller.py
|   |-- lqg_controller.py
|   |-- smith_predictor.py
|   |-- feedforward.py
|   +-- camera_controller.py
|-- pat/                   # PAT state machine
|-- estimation/            # IMM-EKF state estimator
|-- benchmark/             # Evaluation framework
|-- benchmarks/            # Monte Carlo runner
+-- tests/                 # Automated tests
```

1.2 Installation Procedure

1. Open a command prompt or terminal.
2. Navigate to the HORIZON project & Create & activate the recommended Python virtual env.
3. Install the dependencies listed in `requirements.txt`.
4. Run the supplied smoke-test command before launching the graphical interface.

```
# Step 1: Navigate to project directory
cd d:\Hackathon\SIH2.0\HORIZON\coarse-align-x

# Step 2: Create virtual environment (recommended)
python -m venv venv

# Step 3: Activate virtual environment
venv\Scripts\activate

# Step 4: Install all dependencies
pip install -r requirements.txt

# Step 5: Verify installation
python -m pytest tests/test_sota_upgrade.py tests/test_control_sota_part_e.py -v
```

The supplied installation check is intended as the initial smoke test before normal application use.

The project path shown above is the path supplied in the current HORIZON development environment. When the project is located elsewhere, replace the `cd` path with the actual HORIZON project directory.

1.3 First Launch

Launch the graphical application using:

```
python main.py --gui
```

After successful launch, the HORIZON instrument console opens on the **Mission Control** screen.

The documented display target for the current interface is a minimum resolution of 1920×1080.

User Note The normal operator workflow begins at Mission Control. From there, configure the experiment, launch the runtime, observe the Live Tracking screen, inspect Track Diagnostics, apply disturbance conditions when required, and proceed to the benchmarking workflow.

1.4 Command-Line Quick Start

HORIZON also supports headless execution through `main.py`. The following command examples are retained from the supplied release documentation.

1.4.1 Figure-8 Run with Telemetry Export

```
python main.py --trajectory figure8 --duration 10.0 \\  
  --export-csv data/ground_truth/fig8.csv \\  
  --export-json data/ground_truth/fig8.json
```

1.4.2 Circular Run with Custom Seed

```
python main.py --trajectory circular --seed 12345 --duration 10.0
```

1.4.3 Severe Disturbance Preset

```
python main.py --trajectory figure8 --preset SEVERE --duration 15.0
```

1.4.4 Custom YAML Configuration

```
python main.py --config configs/default.yaml
```

1.5 Basic CLI Arguments

Table 3: Command-line arguments documented for HORIZON operation.

Argument	Type	Default	Description
-gui	flag	off	Launch the PySide6 graphical interface
-config	str	configs/default. yaml	Specify the YAML configuration file
-trajectory	str	straight	Select straight / circular / figure8 / random / spiral / sinusoidal
-seed	int	42	Specify the random seed
-duration	float	10.0	Simulation duration in seconds
-preset	str	NOMINAL	Select NOMINAL / DIFFICULT / SEVERE / RECOVERY / ADVERSARIAL
-export-csv	str	data/ground_truth/ experiment.csv	Specify the CSV export path
-export-json	str	None	Specify the JSON export path

1.6 Runtime Output Locations

The current HORIZON documentation specifies the following runtime directories.

Table 4: Runtime output directories used by HORIZON.

Directory	Contents
logs/	simulation.log — timestamped event log
data/ground_truth/	Exported CSV / JSON telemetry
results/	Benchmark JSON results
reports/	Automatically generated HTML / PDF validation reports
runs/	Per-run experiment metadata

User Note The output paths above are the documented locations associated with the current HORIZON release. During an evaluation session, these directories provide the operator with the generated telemetry, benchmark results, reports, and run-specific metadata.

1.7 First-Launch Verification

Before starting a normal tracking experiment, verify the following:

Check	Expected Condition	Status
01	Python environment is active	
02	Dependencies install without error	
03	Supplied smoke-test command completes	
04	HORIZON GUI launches successfully	
05	Mission Control screen is visible	
06	Project output directories are available when required	

1.8 Application Shell Overview

HORIZON uses a dark **PySide6 instrument-suite** presentation designed for continuous telemetry observation and operational analysis. The global navigation remains visible while the active workstation changes between mission setup, live tracking, diagnostics, disturbance testing, and benchmarking.

The application shell is organized into three persistent areas:

1. **Application Header** — provides the current system state, processing frequency, experiment clock, and execution status.
2. **Workstation Navigation** — provides direct access to the five operational workstations.
3. **Active Workstation Area** — displays the controls, visualizations, telemetry, and operational information associated with the selected workstation.

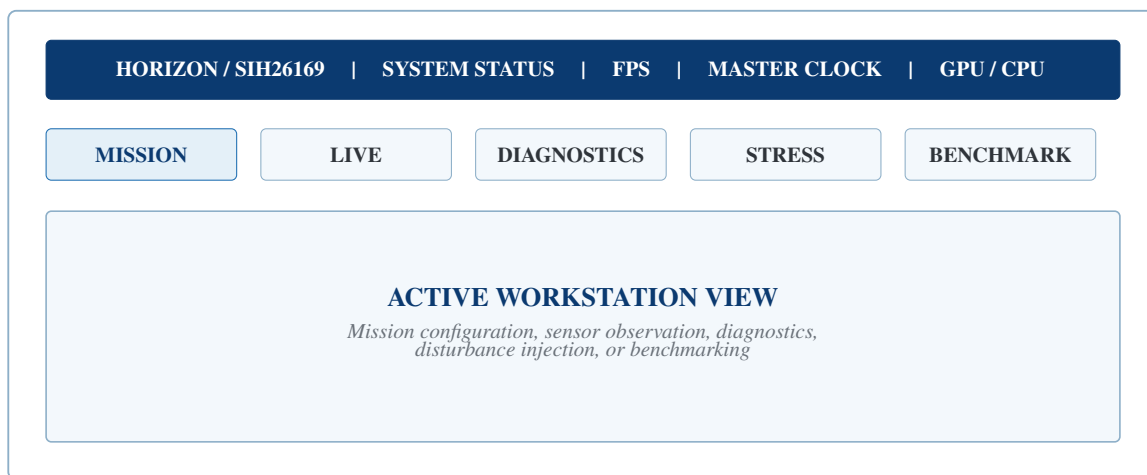


Figure 1: HORIZON application shell with persistent status monitoring, workstation navigation, and the active workspace.

1.9 Global Workstation Navigation

The HORIZON interface provides five primary workstations. Navigation changes the visible workstation while keeping the current application session available for continued operation.

Table 5: HORIZON workstation navigation and operational purpose.

Key	Workstation	Primary Use	Operator Function
1	Mission Control	Scenario setup	Select the experiment, configure the main operating values, and preview the intended run before execution.
2	Live Tracking	Real-time observation	Observe the sensor view, tracking state, target position, and active pointing response during execution.
3	Track Diagnostics	Track analysis	Inspect tracking error, estimator-related diagnostics, and signal information during or after a run.
4	Disturbance Room	Stress testing	Apply disturbance conditions and observe the response of the tracking system.
5	Benchmarking	Performance evaluation	Run and inspect the benchmark and evaluation workflow.

1.10 Application Header

The application header remains visible across the HORIZON workstations. It provides a compact status summary so that the operator does not need to leave the current workspace to determine whether the application is executing normally.

The header reports four main categories of information:

1. **System Status** — identifies the current engine state as IDLE, RUNNING, PAUSED, or STOPPED.
2. **FPS Meter** — reports the current frame-processing frequency.
3. **Master Clock** — displays the experiment time associated with the current execution.
4. **Execution Status** — indicates whether processing is being performed using the available GPU/CPU execution path.

SYSTEM STATUS IDLE / RUNNING PAUSED / STOPPED	FPS METER Current processing frequency	MASTER CLOCK Experiment time during execution	EXECUTION GPU / CPU processing status
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Figure 2: Operator-facing information exposed through the HORIZON application header.

1.11 Status Indicators and Visual Hierarchy

HORIZON uses a consistent visual language to distinguish normal operation, active tracking, degraded operation, and loss or error conditions.

Table 6: Primary HORIZON interface color tokens and their operational meaning.

UI Role	Token	Operational Meaning
Primary Background	#0A0E1A	Midnight-black application field.
Raised Panel	#13192B	Workstation surfaces and information panels.
Hairline Border	#1E293B	Structural separation between interface elements.
Active State	#00F0FF	Active navigation or diagnostic state.
Confirmed Lock	#00FF66	Positive tracking indication.
Degraded	#FFB000	Warning or reduced-quality state.
Loss / Error	#FF2A6D	Failure or critical indication.

IDLE	TRACK	CONFIRMED	DEGRADED	REACQUIRE
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Figure 3: Representative operator-facing tracking status indicators.

User Note

Status color provides a rapid visual cue. The corresponding text state should always be read as the authoritative operational indicator.

1.12 Global Hotkeys & Navigation Shortcuts

Keyboard shortcuts provide direct access to common execution and navigation actions. They are intended to reduce unnecessary movement between interface controls during an active experiment.

Table 7: Global keyboard shortcuts supplied for HORIZON operation.

Key	Action	Operational Meaning
Space	Play / Pause	Toggle execution.
R	Reset	Return the clock and internal tracking state to the initial condition.
1	Mission Control	Open scenario setup and preview.
2	Live Workstation	Open the real-time sensor and PAT view.
3	Track Diagnostics	Open error, covariance, and signal diagnostics.
4	Disturbance Room	Open the real-time disturbance and stress workspace.
5	Benchmarking	Open the benchmark and evaluation workspace.
Esc	Emergency Stop	Abort the active run immediately.

1.13 Operator Navigation Logic

The five workstations are intended to support a continuous operator workflow:



Figure 4: Operator navigation flow across the five HORIZON workstations.

This navigation sequence reflects the intended operational flow of the interface: configure the experiment, observe the execution, inspect the resulting behavior, apply stress conditions when required, and proceed to formal evaluation.

The operator may also move directly between workstations using the navigation controls or the corresponding keyboard shortcuts.

User Note

Changing the visible workstation does not by itself represent a new experiment. Navigation changes the operator's view of the system, while the current execution context remains available to the application.

WORKSTATION WALKTHROUGHS

2 Workstation Walkthroughs

2.1 Screen 1 — Mission Control

The **Mission Control** screen is the pre-run workspace for selecting an experiment, resolving configuration, and previewing the target motion before execution.



Figure 5: Mission Control organization: scenario selection, configuration, preview, and resolved parameters.

2.1.1 Component Breakdown

1. **Scenario Gallery:** a grid of pre-configured mission scenarios. Each scenario tile communicates the physical environment and difficulty class.
2. **Configuration Form:** provides user overrides for master random seed, simulation duration, initial target range, and camera FOV.
3. **Experiment Preview:** displays the configured ground-truth trajectory projected into camera azimuth/elevation coordinates before execution.
4. **Resolved Summary:** reports the final parameter set that the experiment will use.

2.1.2 Scenario Difficulty Classes

Table 8: Scenario difficulty classes supplied for the Mission Control workspace.

Class	Operational Description
NOMINAL	Clear, stable tracking conditions.
DIFFICULT	Increased atmospheric turbulence and platform vibration.
SEVERE	Dense cloud/opacity effects and elevated thermal noise.
RECOVERY	Target enters from outside the active viewing region and requires acquisition/recovery.
ADVERSARIAL	Evasive target behavior combined with reflective distractors.

2.2 Screen 2 — Live Tracking Workstation

The **Live Tracking Workstation** is the primary real-time observation screen during an experiment. It combines the sensor image, PAT state, world geometry, and current telemetry.

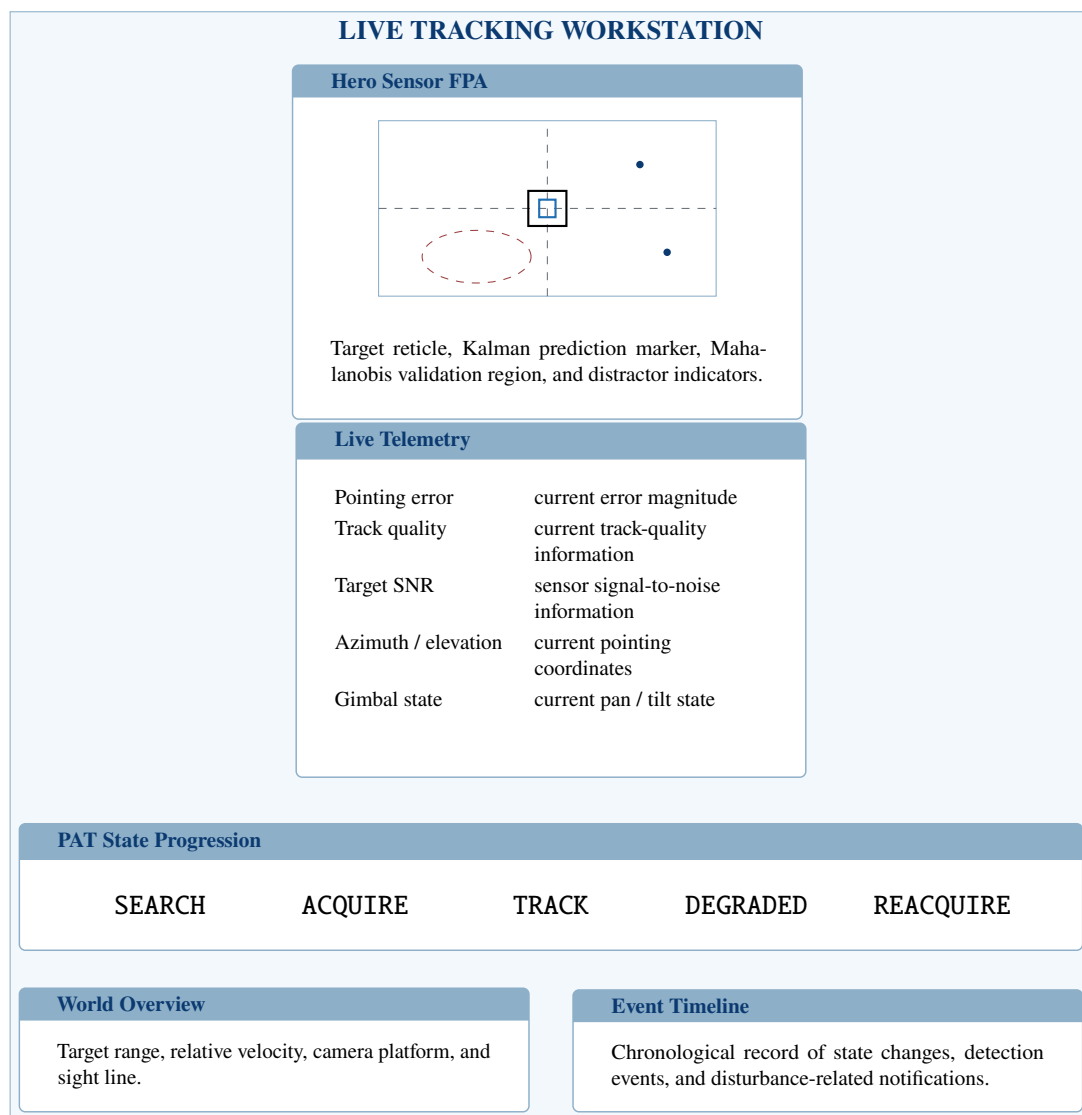


Figure 6: Live Tracking Workstation: sensor observation, telemetry, PAT state progression, and world context.

2.2.1 Component Breakdown

1. **Hero Sensor FPA View:** displays the current sensor image and overlays the extracted centroid, prediction marker, validation region, and distractor indicators.
2. **PAT State Stepper:** shows the active state in the SEARCH → ACQUIRE → TRACK → DEGRADED → REACQUIRE lifecycle.
3. **Live Telemetry:** reports pointing error magnitude, estimated velocity, gimbal pan/tilt state, and sensor SNR.
4. **World Overview:** provides the spatial relation between the camera platform, target vector, and pointing direction.
5. **Event Timeline:** records state changes, detection events, and disturbance-related notifications in chronological order.

2.2.2 Interpreting the Live View

The sensor overlay and telemetry should be interpreted together. A valid centroid with a high-quality state estimate should be accompanied by a stable PAT state and a consistent pointing response.

A loss of detection may first appear as a reduction in track quality before the PAT manager transitions to a degraded or recovery state.

2.3 Screen 3 — Track Diagnostics Workstation

The **Track Diagnostics Workstation** is intended for engineering inspection rather than basic operation. It exposes tracking error, filter consistency, covariance behavior, and frequency-domain diagnostics.

2.3.1 Analytics Graphs

1. **Pointing Error vs. Time:** shows instantaneous tracking error history relative to the configured evaluation threshold.
2. **Tracking Error CDF:** shows the cumulative distribution of observed errors.
3. **Power Spectral Density (PSD):** decomposes platform jitter and observed response into frequency components.
4. **Sub-pixel Centroid Residuals:** visualizes residual centroid variation around the estimated target location.

2.3.2 Covariance, Innovation, and Model Probabilities

The covariance display reports the estimator covariance trace, innovation information, and model-probability history.

2.3.3 Kinematic Scope & Export

The kinematic scope displays relative azimuth, elevation, and range.

The Snapshot Exporter provides high-resolution PNG diagnostics plus CSV telemetry and JSON state traces for later analysis.

User Note

Use Track Diagnostics when a run behaves unexpectedly. The purpose is not merely to show that the beacon was lost, but to identify whether the observed behavior originated in measurement quality, state uncertainty, dynamics, or pointing response.

2.4 Screen 4 — Disturbance Experiment Room

The **Disturbance Experiment Room** allows the operator to increase physical/environmental stress during an active run and observe how the tracking system responds.

2.4.1 Live Diagnostic Overlays

- **Jitter PSD Overlay:** shows the evolving spectral distribution of the injected structural vibration.
- **System Response Sensitivity Panel:** presents a heatmap of tracking error as disturbance parameters change.

2.5 Screen 5 — Scientific Benchmarking Workstation

The **Scientific Benchmarking Workstation** executes automated Monte Carlo evaluations and summarizes algorithmic performance across the configured scenario set.

2.5.1 Monte Carlo Suite Controller

The **Monte Carlo Suite Controller** configures multi-seed execution using selectable batch sizes, including $N = 10$, 50, 100, or 500 seeds per scenario, with parallel execution where supported by the implementation.

2.5.2 Algorithm Comparison

The comparison table presents side-by-side performance for:

1. Classical Thresholding + Standard EKF,
2. CNN Detector + Standard EKF,
3. HORIZON Hybrid Fusion + IMM-EKF.

The comparison is intended to use consistent experimental conditions so that observed differences can be attributed to the evaluated pipeline rather than to different scenario inputs.

2.5.3 Statistical Distribution Plots

The distribution view visualizes the distribution of:

- tracking error,
- lock duration,
- time-to-first-acquisition.

Violin and box plots expose spread and outliers that a single mean value can hide.

2.5.4 Failure Intelligence Taxonomy

The failure-analysis view categorizes observed failure events into operational classes such as:

- loss of lock during severe atmospheric fading,
- false association with a high-brightness distractor,
- gimbal rate saturation during high-velocity target motion.

2.5.5 Automated Report Export

The benchmark workstation can generate publication-oriented HTML/PDF benchmark dossiers containing statistical tables, confidence intervals, and compliance-oriented result summaries.

User Note

Benchmark results should always be interpreted together with the configuration and seed information. A single successful run is not a substitute for a repeatable experiment.

OPERATIONAL WORKFLOWS & SOPs

This section defines the normal operator procedures for preparing, executing, stressing, benchmarking, and reviewing a HORIZON experiment.

The procedures are written as execution checklists rather than algorithm descriptions. The operator should use the procedures in combination with the workstation descriptions given in Section 3.



Figure 7: Primary HORIZON operating sequence from environment validation to benchmark execution.

3 Workflows

3.1 SOP-01 — First-Time System Setup & Environment Validation

Purpose: Verify that the HORIZON execution environment is installed correctly before starting a normal tracking experiment.

SOP-01 CHECKLIST

1. Open a command prompt or terminal.
2. Navigate to the HORIZON project directory.
3. Create the recommended Python virtual environment.
4. Activate the virtual environment.
5. Install the dependencies from `requirements.txt`.
6. Run the supplied installation verification command.
7. Confirm that the graphical application launches successfully.

3.1.1 Installation Command Sequence

```
cd d:\Hackathon\SIH2.0\HORIZON\coarse-align-x

python -m venv venv

venv\Scripts\activate

pip install -r requirements.txt

python -m pytest tests/test_sota_upgrade.py tests/test_control_sota_part_e.py -v
```

The supplied installation check is intended as the initial smoke test before normal application use. The project path above is the path supplied in the current HORIZON development environment and should be replaced when the project is located elsewhere. This procedure is consistent with the installation workflow already documented.

3.1.2 Launch Verification

After installation, launch the graphical application:

```
python main.py --gui
```

After successful launch, the HORIZON instrument console opens on the **Mission Control** screen.

The documented display target for the current interface is a minimum resolution of 1920×1080.

3.2 SOP-02 — Executing a Standard Tracking Mission

Purpose: Execute a normal HORIZON tracking experiment using the Mission Control and Live Tracking workstations.

SOP-02 CHECKLIST

1. Open **Mission Control**.
2. Select the required scenario from the scenario gallery.
3. Review the available environment and difficulty information associated with the selected scenario.
4. Configure the supported experiment values, including seed, duration, target range, and camera FOV.
5. Review the **Experiment Preview**.
6. Confirm the **Resolved Summary**.
7. Start the experiment.
8. Move to the **Live Tracking Workstation**.
9. Observe the sensor view, PAT state, telemetry, and world overview during execution.
10. Inspect the event timeline for state changes, detection events, or disturbance notifications.

3.2.1 Pre-Run Review

Before starting the run, confirm:

Check	Operator Question	Expected Action
01	Is the intended scenario selected?	Continue after confirming the scenario.
02	Do the configured values match the intended experiment?	Review the configuration fields.
03	Does the preview show the intended target motion?	Confirm before starting execution.
04	Is the resolved summary consistent with the requested setup?	Proceed only after review.

The original Mission Control material specifically defines the scenario gallery, configuration form, experiment preview, and resolved summary as the four principal operator areas. :contentReference[oaicite:9]index=9

3.3 SOP-03 — Real-Time Disturbance Injection

Purpose: Apply controlled environmental or sensor stress during an active experiment and observe the response of the tracking system.

SOP-03 CHECKLIST

1. Start the intended tracking experiment.
2. Open the **Disturbance Experiment Room**.
3. Select the disturbance control that is to be varied.
4. Adjust the disturbance level through the available control.

5. Observe the live tracking response.
6. Monitor the PAT state, tracking behavior, and diagnostic overlays.
7. Record the resulting behavior through the available telemetry and run outputs.

3.3.1 Documented Disturbance Controls

Table 9: Documented disturbance controls available in the HORIZON stress workspace.

Control	Documented Range	Operational Effect
Platform jitter amplitude	0.0–50.0 px	Changes vibration displacement amplitude.
Platform jitter frequency	1.0–50.0 Hz	Changes vibration frequency.
Atmospheric turbulence C_n^2	10^{-17} – 10^{-12}	Changes refractive-index fluctuation severity.
Thermal noise σ_n	0.0–30.0 DN	Changes Gaussian readout noise.
Impulse noise rate p	0.0%–10.0%	Changes salt-and-pepper corruption probability.

3.3.2 Operator Observation Sequence

The operator should observe the disturbance and tracking displays together:



This procedure is intentionally observational: the purpose of the stress workspace is to expose system behavior under changed conditions rather than conceal degradation or recovery events.

3.4 SOP-04 — Running the Multi-Seed Monte Carlo Suite

Purpose: Execute repeatable batch evaluation runs and retain the resulting benchmark artifacts.

The supplied benchmark documentation describes a multi-seed evaluation process, while the CLI documentation provides the corresponding headless benchmark command.

SOP-04 CHECKLIST

1. Confirm the benchmark configuration before starting the batch.
2. Use an explicit configuration file when reproducibility is required.
3. Set an explicit seed for runs where a controlled starting condition is required.
4. Launch the benchmark suite through the Benchmarking workstation or the documented command-line interface.
5. Monitor batch progress and execution status.
6. Retain the generated JSON/CSV/PDF outputs associated with the run.
7. Record the configuration and seed information together with the resulting benchmark artifacts.

3.4.1 Headless Benchmark Execution

The documented command-line syntax is:

```
python main.py [FLAGS] [OPTIONS]
```

The supplied benchmark example is:

```
python main.py --benchmark --threads 16 \
  --export-json results/suite_results.json
```

A severe-disturbance sweep is documented as:

```
python main.py --trajectory figure8 --preset SEVERE --seed 9999 \
  --duration 30.0 \
  --export-csv data/ground_truth/severe_sweep.csv
```

The supplied CLI documentation also gives:

```
python -m pytest tests/ -q
```

for a headless regression-test pass. :contentReference[oaicite:12]index=12

3.4.2 Benchmark Artifact Handling

The operator should retain:

Artifact	Purpose
Configuration file	Records the experiment settings.
Seed information	Identifies the controlled random starting condition.
CSV telemetry	Provides frame-level exported telemetry where requested.
JSON result	Provides structured benchmark output.
PDF report	Provides the generated benchmark report where enabled.

For reproducibility, the supplied User Manual guidance recommends retaining the configuration, random seed, software revision, and output paths together with the experiment artifacts. :contentReference[oaicite:13]index=13

3.5 SOP-05 — External Evaluator Video / HIL Data Input

Purpose: Use an external recorded video source for the coarse-pointing evaluation path.

The HORIZON documentation establishes a dual-mode ingestion design: the system can operate with the synthetic execution path or ingest external MP4/AVI video, with the recorded video connected directly to the coarse-pointing path while the physical PTZ actuation path is bypassed for observation-only processing. :contentReference[oaicite:14]index=14

SOP-05 CHECKLIST

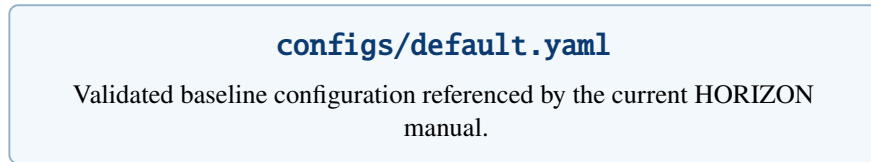
1. Open the HORIZON live-tracking workflow.
2. Select the documented external-video input mode.
3. Provide the evaluator-supplied MP4 or AVI file.
4. Start processing through the external-video path.
5. Observe the sensor/track outputs and tracking telemetry.
6. Review the resulting centroiding and performance outputs.
7. Export the generated telemetry or benchmark artifacts required for the evaluation record.

CONFIGURATION DICTIONARY

4 Configuration Overview

HORIZON configuration management is governed by the validated configuration schemas implemented in `simulator/core/config.py`. System parameters can be loaded from YAML files and, where supported by the application, overridden through command-line options. :contentReference[oaicite:2]index=2

The primary configuration file referenced by the supplied manual material is:



4.1 Configuration Hierarchy

The supplied configuration is divided into six major groups:

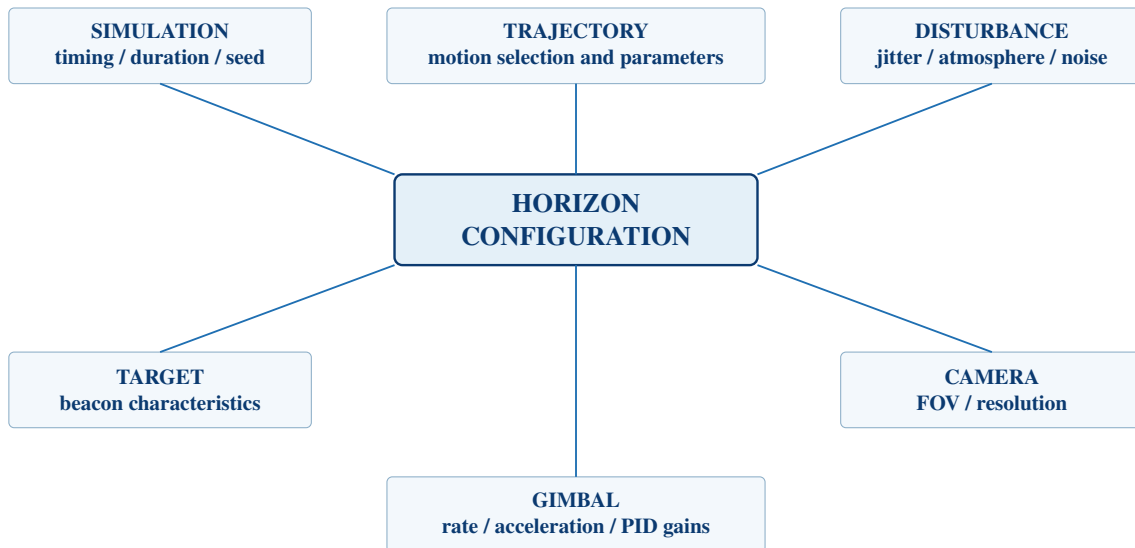


Figure 8: Configuration hierarchy of the six groups supplied in the HORIZON configuration reference.

Table 10: Configuration groups and their operational roles.

Group	Key Parameters	Operational Purpose
Simulation	frequency, duration, seed	Controls master timing, experiment length, and reproducibility.
Trajectory	type, amplitudes, velocity, angular velocity	Selects and parameterizes target motion.
Disturbance	preset, jitter, atmosphere, noise	Defines environmental and sensor degradation.
Target	size, intensity, wavelength	Defines beacon appearance and optical characteristics.
Camera	FOV, resolution	Defines active sensor geometry.
Gimbal	rate, acceleration, PID gains	Defines pointing-control and actuator constraints.

4.2 Master YAML Configuration

The following configuration block reproduces the supplied HORIZON master configuration schema and values from the configuration reference.

```
# HORIZON System Master Configuration Schema
```

```
simulation:
  frequency_hz: 100.0
  duration_seconds: 10.0
  seed: 42

trajectory:
  type: "figure8"
  figure8:
    amplitude_x: 220.0
    amplitude_y: 140.0
    angular_velocity: 0.60
  straight:
    velocity_x: 120.0
    velocity_y: 40.0

disturbance:
  preset: "SEVERE"
  jitter:
    enabled: true
    amplitudes: [5.0, 12.0, 20.0]
    frequencies: [5.0, 12.5, 27.8]
  atmosphere:
    enabled: true
    cn2_structure: 1.0e-13
    rytov_variance: 0.25
  noise:
    readout_std: 8.5
    impulse_rate: 0.01

target:
  size_px: 10.0
  intensity: 255.0
  wavelength_nm: 850.0

camera:
  fov_deg: 10.0
  resolution: [256, 256]

gimbal:
  rate_limit_deg_s: 20.0
  accel_limit_deg_s2: 300.0
  kp: 12.5
  ki: 0.50
  kd: 0.85
```

The values above are reproduced from the supplied Part VI configuration material. The source explicitly notes that some defaults differ elsewhere in the HORIZON documentation, including camera resolution/FOV and trajectory defaults. Before submission, the configuration documented here should match the configuration.

4.3 Simulation Parameters

The simulation group contains the master execution settings.

Table 11: Simulation configuration parameters supplied in the HORIZON configuration reference.

Parameter	Value	Type	Operational Meaning	Documented Range
frequency_hz	100.0	Float	Master simulation frequency.	Not specified in supplied configuration documentation.
duration_seconds	10.0	Float	Simulation run duration in seconds.	Not specified in supplied configuration documentation.
seed	42	Integer	Master random seed used for controlled, repeatable execution.	Not specified in supplied configuration documentation.

4.4 Trajectory Parameters

The trajectory group selects the target-motion profile and contains the documented parameters for the supplied Figure-8 and Straight configurations.

Table 12: Trajectory configuration parameters supplied in the HORIZON configuration reference.

Parameter	Value	Type	Operational Meaning	Documented Range / Choices
trajectory.type	figure8	String	Selects the active target-motion type.	straight, circular, figure8, random, spiral, sinusoidal
figure8.amplitude_x	220.0	Float	X-axis oscillation amplitude in the supplied Figure-8 profile.	Not specified
figure8.amplitude_y	140.0	Float	Y-axis oscillation amplitude in the supplied Figure-8 profile.	Not specified
figure8.angular_velocity	0.60	Float	Fundamental Figure-8 angular-velocity parameter.	Not specified
straight.velocity_x	120.0	Float	X-axis target velocity for the supplied Straight profile.	Not specified
straight.velocity_y	40.0	Float	Y-axis target velocity for the supplied Straight profile.	Not specified

User Note

Only the Figure-8 and Straight parameter blocks are explicitly shown in the supplied master YAML example. The configuration reference identifies additional selectable trajectory types, but does not provide their complete parameter blocks in the reproduced schema. They are therefore not expanded here.

4.5 Disturbance Parameters

The disturbance group defines the environmental and sensor conditions represented in the supplied configuration.

4.5.1 Disturbance Preset

Parameter	Value	Operational Meaning
disturbance.preset	SEVERE	Selects the configured disturbance condition.

The supplied documentation identifies the following preset names:

NOMINAL DIFFICULT SEVERE RECOVERY ADVERSARIAL

4.5.2 Jitter Configuration

Parameter	Supplied Value	Type	Operational Meaning
jitter.enabled	true	Boolean	Enables the configured jitter disturbance.
jitter.amplitudes	[5.0, 12.0, 20.0]	List	Supplied vibration-amplitude values in pixels.
jitter.frequencies	[5.0, 12.5, 27.8]	List	Supplied vibration-frequency values in Hz.

4.5.3 Atmospheric Configuration

Parameter	Supplied Value	Type	Operational Meaning
atmosphere.enabled	true	Boolean	Enables the configured atmospheric disturbance.
atmosphere.cn2_structure	1.0×10^{-13}	Float	Supplied refractive-index structure parameter.
atmosphere.rytov_variance	0.25	Float	Supplied log-amplitude variance parameter.

4.5.4 Sensor Noise Configuration

Parameter	Supplied Value	Type	Operational Meaning
noise.readout_std	8.5	Float	Supplied Gaussian readout-noise standard deviation in DN.
noise.impulse_rate	0.01	Float	Supplied salt-and-pepper corruption probability.

4.6 Target Parameters

The target group defines the supplied beacon characteristics.

Parameter	Supplied Value	Type	Operational Meaning
target.size_px	10.0	Float	Supplied beacon Gaussian-profile size parameter in pixels.
target.intensity	255.0	Float	Supplied peak photon intensity, on the documented 0–255 DN scale.
target.wavelength_nm	850.0	Float	Supplied beacon optical wavelength in nanometres.

4.7 Camera Parameters

The camera group defines the active sensor geometry.

Parameter	Supplied Value	Type	Operational Meaning
camera.fov_deg	10.0	Float	Supplied camera field of view in degrees.
camera.resolution	[256, 256]	Integer pair	Supplied focal-plane-array dimensions in pixels.

The supplied SIH26169 problem statement and other HORIZON material contain camera values that differ from the values reproduced in this configuration example. This section intentionally reproduces the supplied Part VI configuration rather than silently changing it. Before the final manual release.

4.8 Gimbal and PID Parameters

The gimbal group defines the supplied pointing-rate, acceleration, and PID parameters.

Parameter	Supplied Value	Type	Operational Meaning
<code>gimbal.rate_limit_deg_s</code>	20.0	Float	Supplied maximum slew-rate parameter in degrees per second.
<code>gimbal.accel_limit_deg_s2</code>	300.0	Float	Supplied maximum slew-acceleration parameter in degrees per second squared.
<code>gimbal.kp</code>	12.5	Float	Supplied proportional control gain.
<code>gimbal.ki</code>	0.50	Float	Supplied integral control gain.
<code>gimbal.kd</code>	0.85	Float	Supplied derivative control gain.

4.9 GUI Configuration Fields Explicitly Documented

The Mission Control material explicitly identifies the following operator-facing configuration controls:

GUI Field	Operator Function	Configuration Reference Status
Master Random Seed	Sets the experiment seed used for the run.	A corresponding <code>simulation.seed</code> field is documented.
Simulation Duration	Sets the requested experiment duration.	A corresponding <code>simulation.duration_seconds</code> field is documented.
Initial Target Range	Provides the documented target-range configuration control.	A corresponding YAML field is not identified in the supplied Part VI configuration excerpt.
Camera FOV	Provides the camera field-of-view control.	A corresponding <code>camera.fov_deg</code> field is documented.

User Note

The table above intentionally distinguishes between GUI controls explicitly described in the workstation material and YAML fields explicitly shown in the configuration reference.

4.10 Operator Procedure for Editing Configuration

When a configuration file must be changed, use the following operator sequence:

1. Identify the configuration group containing the parameter to be changed.
2. Change only the required parameter value.
3. Preserve the YAML hierarchy and parameter spelling.
4. Save the configuration file.
5. Load the configuration through the documented HORIZON workflow.
6. Verify the resolved values in Mission Control before starting the experiment.
7. Retain the configuration file together with the resulting run artifacts.

Do not assume that a value appearing in one HORIZON document is automatically the active value of the released executable. The supplied Part VI source explicitly warns that defaults differ in some parts of the documentation and that the final manual should match the configuration actually loaded by the executable.

HEADLESS CLI & AUTOMATION GUIDE

5 Command-Line Operation Overview

HORIZON supports command-line execution for repeatable experiments, batch benchmarking, configuration sweeps, and automated regression testing. The command-line entry point is `main.py`.

Basic command syntax

```
python main.py [FLAGS] [OPTIONS]
```

The headless interface is intended for execution without relying on the interactive workstation display. It uses the documented configuration, trajectory, preset, seed, duration, benchmark, and export options provided by HORIZON.

5.1 CLI Execution Model

A headless run can be treated as a four-stage operator workflow:



Figure 9: Operator-oriented headless execution flow for HORIZON.

5.2 Complete CLI Options

The following command-line options reproduce the supplied HORIZON CLI reference.

Table 13: Command-line options supplied for the HORIZON application.

Flag	Type	Default	Description
<code>-gui</code>	Flag	False	Launch interactive PySide6 graphical user interface.
<code>-config</code>	File path	<code>configs/default.yaml</code>	Load a custom YAML configuration file.
<code>-trajectory</code>	String	<code>figure8</code>	Select <code>straight</code> , <code>circular</code> , <code>figure8</code> , <code>random</code> , <code>spiral</code> , or <code>sinusoidal</code> .
<code>-preset</code>	String	<code>NOMINAL</code>	Select <code>NOMINAL</code> , <code>DIFFICULT</code> , <code>SEVERE</code> , <code>RECOVERY</code> , or <code>ADVERSARIAL</code> .
<code>-seed</code>	Integer	42	Master random seed used for reproducible execution.
<code>-duration</code>	Float	10.0	Simulation duration in seconds.
<code>-export-csv</code>	File path	None	Export frame-by-frame telemetry.
<code>-export-json</code>	File path	None	Export a structured benchmark summary.
<code>-export-pdf</code>	File path	None	Generate a PDF benchmark report.
<code>-benchmark</code>	Flag	False	Execute the full benchmark suite.
<code>-threads</code>	Integer	8	Number of parallel worker threads.

User Note

The defaults above reproduce the supplied Part VI command-line reference. Before final release of the manual, verify them against the executable used for the evaluation build.

5.3 Standard Headless Command Profiles

The following command profiles are reproduced from the supplied HORIZON operational material. :contentReference[oaicite:3]index=3

5.3.1 Benchmark Suite

```
python main.py --benchmark --threads 16 \
  --export-json results/suite_results.json
```

This command requests benchmark-suite execution with parallel worker threads and JSON result export.

5.3.2 Severe-Disturbance Sweep

```
python main.py --trajectory figure8 --preset SEVERE --seed 9999 \
  --duration 30.0 \
  --export-csv data/ground_truth/severe_sweep.csv
```

This command executes the supplied Figure-8 / SEVERE example with the specified seed, duration, and CSV telemetry output path.

5.3.3 Headless Regression Test

```
python -m pytest tests/ -q
```

This command executes the supplied full-test invocation in quiet mode.

5.3.4 Custom Configuration Run

```
python main.py --config configs/default.yaml
```

This command launches a run using the supplied YAML configuration file.

5.4 Configuration and Command-Line Overrides

For an operator, the practical distinction is:

Approach	Use	Operator Record
YAML configuration	Use a named configuration file containing the experiment settings.	Retain the configuration file with the run artifacts.
CLI option	Use a supported command-line option to select or override an execution value.	Record the complete command used for the run.
Combined	Use the YAML configuration together with supported CLI options.	Retain both the configuration file and the command line.

User Note

For reproducibility, the supplied documentation recommends an explicit configuration file and an explicit seed rather than relying on implicit application defaults. Record the command, configuration file, software revision, and output paths with the experiment artifacts. :contentReference[oaicite:5]index=5

5.5 Batch Execution Checklist

Before launching a long or automated headless run, use the following operator checklist.

HEADLESS RUN CHECKLIST

1. Confirm the intended configuration file.
2. Confirm the intended trajectory and preset when those options are supplied on the command line.
3. Set an explicit seed when reproducibility is required.
4. Confirm the requested duration.
5. Confirm every requested export path.
6. Confirm the benchmark flag before launching a benchmark suite.
7. Confirm the requested worker-thread count when `-threads` is used.
8. Retain the exact command used for the experiment record.

5.6 Automated Regression Use

The supplied CLI material includes a headless pytest invocation for regression testing:

```
python -m pytest tests/ -q
```

This provides a command-line verification path without requiring the interactive GUI.

The supplied User Manual does not prescribe a particular external CI service, job scheduler, shell framework, or cluster platform. Therefore, no additional automation framework is specified here.

SCENARIO MATRIX

7 Scenario Library Overview

HORIZON provides the supplied 13-profile mission library for repeatable operator selection through the Mission Control workflow. The profiles are intended to cover nominal operation, motion diversity, environmental degradation, target loss/recovery, clutter, and user-defined experiments.

The scenario library is an operator selection mechanism: the evaluator selects a profile, reviews its operating condition, and then uses the Mission Control preview and resolved summary before execution.

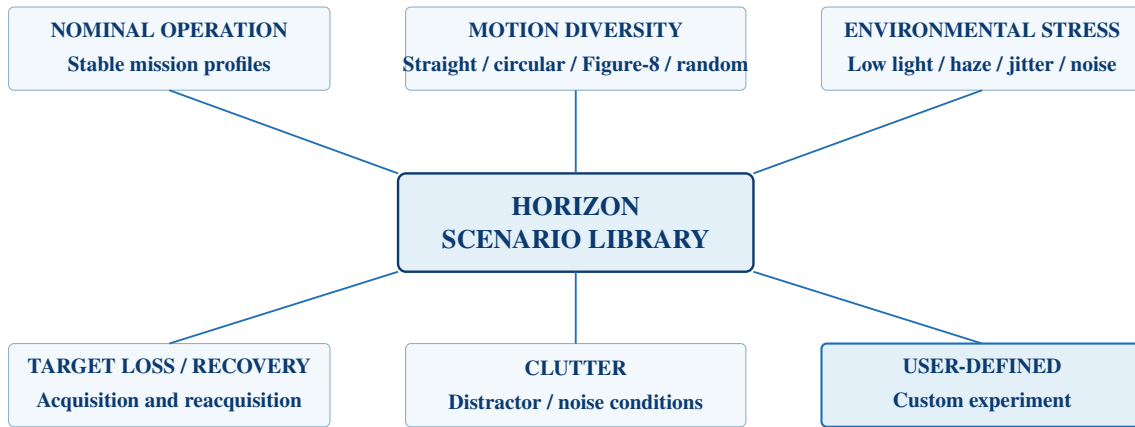


Figure 1: Operator view of the coverage represented by the supplied HORIZON 13-profile scenario library.

7.1 Complete 13-Profile Scenario Matrix

Table 1: HORIZON 13-profile scenario matrix supplied in the configuration and operations material.

ID	Mission Title	Operating Regime	Primary Condition	Difficulty
01	Nominal Acquisition	LEO-to-Ground	Nominal	Nominal
02	Straight Track	LEO Horizon	Nominal	Nominal
03	Circular Track	GEO Inter-sat	Vacuum	Nominal
04	Figure-8 Track	MEO Relay	Moderate disturbance	Moderate
05	Random Track	Rendezvous	Dynamic scintillation	Moderate
06	Low Light	Eclipse ingress	Low photon flux	Moderate
07	Fog / Haze	Downlink	Aerosol haze	Moderate
08	Camera Jitter	Reaction-wheel environment	Platform jitter	Moderate
09	Platform Motion	Kinematic slew	Severe motion	Severe
10	Multi-Distractor	Debris field	Severe noise/clutter	Severe
11	Disturbance Recovery	Occultation	Forced target loss	Severe
12	Severe Combined	Solar-blind / adverse	Maximum stress	Adversarial
13	Custom Experiment	User-defined	Custom disturbance vector	Custom

7.2 Scenario Profiles — Motion and Baseline Conditions

Profiles 01–05

1. **01 — Nominal Acquisition:** baseline lock-on verification under clear-sky conditions with the supplied nominal SNR and zero-vibration configuration.
2. **02 — Straight Track:** constant-velocity transit evaluating steady-state tracking performance.
3. **03 — Circular Track:** circular relative motion with centripetal acceleration.
4. **04 — Figure-8 Track:** dual-axis harmonic trajectory emphasizing direction-change and inflection behavior.
5. **05 — Random Track:** stochastic motion using the configured random-acceleration process.

7.3 Scenario Profiles — Environmental and Clutter Conditions

Profiles 06–10

6. **06 — Low Light:** reduced photon flux with increased shot-noise influence.
7. **07 — Fog / Haze:** degraded contrast caused by atmospheric attenuation/scattering.
8. **08 — Camera Jitter:** multi-frequency platform micro-vibration stress.
9. **09 — Platform Motion:** dynamic platform motion with elevated angular activity.
10. **10 — Multi-Distractor:** bright distractors introduced near the true beacon.

7.4 Scenario Profiles — Recovery, Severe Combined, and Custom

Profiles 11–13

11. **11 — Disturbance Recovery:** target-loss event testing prediction and reacquisition.
12. **12 — Severe Combined:** multiple simultaneous adverse conditions.
13. **13 — Custom Experiment:** user-defined parameters for research and sensitivity studies.

7.5 Scenario Selection Workflow

Use the following procedure when selecting a scenario from Mission Control:

1. Open the **Mission Control** workstation.
2. Select the required scenario profile.
3. Review the scenario's operating regime and primary condition in this matrix.
4. Review the available scenario configuration in the Mission Control configuration area.
5. Inspect the **Experiment Preview** to verify the intended target motion and camera viewing geometry.
6. Review the **Resolved Summary** before starting the run.
7. Start the experiment only after the selected scenario and resolved configuration have been checked.



Figure 2: Operator scenario-selection sequence based on the Mission Control workflow.

7.6 Difficulty Class Reference

The Mission Control material defines five operational difficulty classes:

Class	Operational Description
NOMINAL	Clear, stable tracking conditions.
DIFFICULT	Increased atmospheric turbulence and platform vibration.
SEVERE	Dense cloud/opacity effects and elevated thermal noise.
RECOVERY	Target enters from outside the active viewing region and requires acquisition/recovery.
ADVERSARIAL	Evasive target behavior combined with reflective distractors.

User Note

The difficulty class provides a compact indication of the operating condition associated with the selected Mission Control scenario. The scenario matrix should be read together with the actual resolved configuration shown by the application.

7.7 Operator Reference Card

Table 2: Quick-reference guide for selecting the supplied HORIZON scenario profiles.

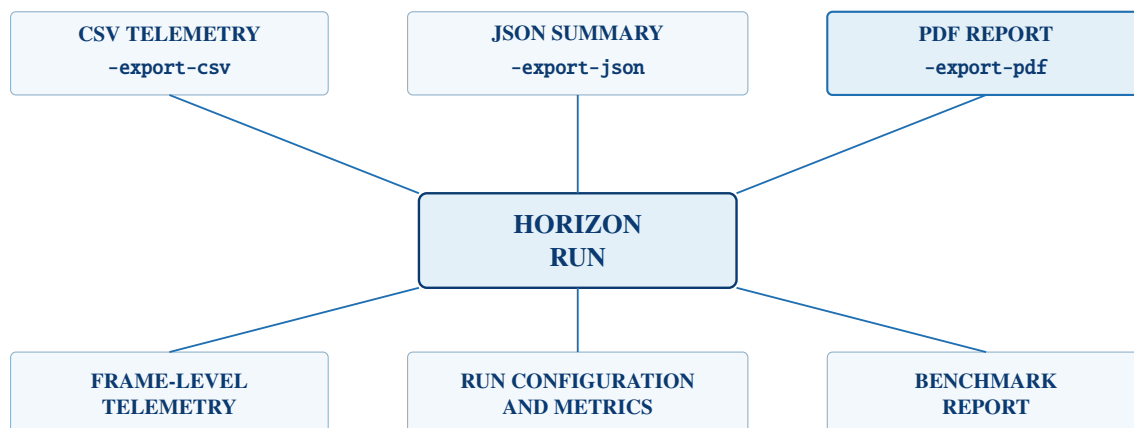
Operator Need	Relevant Profiles	What to Review Before Running
Baseline acquisition	01	Confirm nominal setup and inspect the preview.
Steady motion	02	Review the Straight Track configuration.
Curved motion	03, 04	Inspect the projected trajectory before execution.
Stochastic motion	05	Confirm the selected random-motion setup.
Low-light / atmospheric degradation	06, 07	Review the environmental condition and resolved configuration.
Platform disturbance	08, 09	Check the selected motion/disturbance settings.
Clutter	10	Verify that the selected profile is the intended distractor case.
Recovery testing	11	Confirm that the target-loss/recovery profile is selected.
Combined severe condition	12	Review the resolved severe configuration carefully.
Custom study	13	Verify all user-defined values before execution.

DATA EXPORT & TELEMETRY FORMATS

8 Export Overview

HORIZON provides machine-readable telemetry and structured run summaries through its documented export interfaces.

The supplied export paths are:



8.1 Export Reference

Table 3: Documented HORIZON export options.

Export	CLI Option	Data Level	Operator Use
CSV telemetry	<code>-export-csv</code>	Frame-by-frame	Inspect or process the time history of the run.
JSON summary	<code>-export-json</code>	Run summary	Retain structured configuration and headline metrics.
PDF report	<code>-export-pdf</code>	Report	Generate the documented PDF benchmark report.

The CSV and JSON formats are explicitly documented in the supplied Part VI material. The CLI reference separately lists `-export-pdf` for generating a PDF benchmark report. [:contentReference\[oaicite:3\]index=3](#)
[:contentReference\[oaicite:4\]index=4](#)

8.2 CSV Telemetry Format

Frame-by-frame telemetry is exported in a machine-readable CSV structure. The supplied HORIZON manual defines the following field layout. :contentReference[oaicite:5]index=5

Representative CSV Header

```
frame_idx,
time_s,
target_x_px,
target_y_px,
estimated_x_px,
estimated_y_px,
error_px,
pat_mode,
track_quality,
pan_deg,
tilt_deg,
snr_db
```

The rows above are the representative CSV records supplied in the manual source. They are documentation examples and should not be interpreted as universal measured results.

8.3 CSV Field Reference

Table 4: Primary HORIZON CSV telemetry fields.

Field	Unit / Type	Meaning
frame_idx	Integer	Sequential frame identifier.
time_s	s	Simulation or source-frame time.
target_x_px, target_y_px	px	Reference / ground-truth target coordinates when available.
estimated_x_px, estimated_y_px	px	Estimated beacon coordinates.
error_px	px	Tracking-error magnitude.
pat_mode	Enum	Current PAT state.
track_quality	[0,1]	Composite track-quality value.
pan_deg, tilt_deg	deg	Camera/gimbal angular telemetry.
snr_db	dB	Signal-to-noise indicator.

8.4 Reading a Telemetry Record

A single CSV row represents one recorded frame or time sample.

For example:

```
2,0.020,130.400,128.800,130.392,128.795,0.009,TRACK,0.985,0.024,0.008,24.35
```

The operator can read the supplied example as follows:

Field	Example	Meaning
Frame index	2	Third frame in the supplied example sequence.
Time	0.020 s	Time associated with that record.
Reference position	130.400, 128.800 px	Reference target position when available.
Estimated position	130.392, 128.795 px	Estimated beacon position.
Error	0.009 px	Tracking-error magnitude in the supplied example.
PAT state	TRACK	Current PAT state for the record.
Track quality	0.985	Reported composite track-quality value.
Pan / Tilt	0.024 / 0.008 deg	Camera/gimbal angular telemetry.
SNR	24.35 dB	Signal-to-noise indicator.

User Note

When reference coordinates are unavailable, the corresponding reference fields should not be assumed to represent independently generated ground truth. The supplied documentation explicitly distinguishes reference coordinates from HORIZON's own estimated coordinates.

8.5 Structured JSON Summary Format

The JSON summary provides a compact machine-readable representation of the run configuration and headline metrics. :contentReference[oaicite:6]index=6

Representative JSON Summary

```
{
  "horizon_version": "1.0-RC",
  "timestamp": "2026-09-30T17:00:00Z",
  "scenario": {
    "scenario_id": "figure8",
    "mission_code": "ISRO-PAT-04",
    "master_seed": 42
  },
  "metrics": {
    "total_frames": 1000,
    "rmse_px": 0.478,
    "p95_error_px": 0.835,
    "max_error_px": 1.240,
    "lock_retention_pct": 99.80,
    "mean_snr_db": 18.42,
    "time_to_acquisition_ms": 80.0
  },
  "compliance": {
    "sih26169_passed": true,
    "error_under_3px": true
  }
}
```

8.6 JSON Field Groups

The supplied JSON example is organized into four top-level areas.

Table 5: Top-level groups in the supplied HORIZON JSON summary example.

Group	Fields in Supplied Example	Operator Use
Top-level metadata	horizon_version, timestamp	Identify the software-version label and recorded time.
Scenario	scenario_id, mission_code, master_seed	Identify the run's scenario and seed information.
Metrics	total_frames, rmse_px, p95_error_px, max_error_px, lock_retention_pct, mean_snr_db, time_to_acquisition_ms	Provide the supplied headline performance fields.
Compliance	sih26169_passed, error_under_3px	Provide the supplied compliance indicators for the example.

8.7 PDF Benchmark Report

The command-line reference provides the following option for PDF generation:

```
--export-pdf
```

The supplied CLI documentation describes this option as:

Generate a PDF benchmark report.

No additional universal PDF schema is specified in the supplied User Manual source. Therefore, the exact pages, tables, and visual contents of an exported PDF should be read from the generated report rather than assumed from this manual.

8.8 External-Video Data Handling

For external-video Benchmark-2 runs, reference coordinates supplied by an evaluator must remain distinguishable from HORIZON's estimated coordinates.

The supplied manual explicitly states:

Data Integrity Rule

For external-video Benchmark-2 runs, distinguish reference coordinates supplied by an evaluator from HORIZON's own estimates. An internally generated estimate must never be relabeled as ground truth for error calculation. :contentReference[oaicite:8]index=8

8.9 Recommended Export Record

For a complete operator record, retain the generated export together with the run-identifying information.

Record Item	What to Retain
Run configuration	The configuration used for the run.
Master seed	The seed associated with the run when explicitly configured.
Scenario	Selected scenario identifier / description.
CSV telemetry	Frame-by-frame telemetry export when requested.
JSON summary	Structured run summary when requested.
PDF report	Generated benchmark report when requested.
Command	The complete command used for headless execution, when applicable.

User Note

Keeping the exported data together with the configuration and seed makes it easier to identify exactly which execution produced a given telemetry or summary file.

8.10 Quick Export Reference

CSV

-export-csv
Frame-level telemetry

JSON

-export-json
Run configuration + metrics

PDF

-export-pdf
Benchmark report

This section defines the documented telemetry and export interfaces. Statistical interpretation, benchmark methodology, confidence interval derivations, error-budget analysis, and other research analysis belong in the Technical Report rather than the User Manual.

TROUBLESHOOTING & OPERATOR FAQs

9 Troubleshooting Overview

This section provides the documented operator response for common HORIZON startup, execution, and benchmark-operation issues.

The troubleshooting procedure follows a simple sequence:



9.1 Issue 1 — GUI Fails to Open or UI Scaling Is Misaligned

GUI / DISPLAY SCALING

Symptom

The HORIZON window appears cropped or text becomes difficult to read on a High-DPI Windows display.

Documented Cause

Operating-system display scaling and Qt high-DPI interaction.

Documented Resolution

Launch the application with the supplied Qt scaling environment variable.

9.1.1 Resolution Command

Run the following in Windows PowerShell:

```
$env:QT_AUTO_SCREEN_SCALE_FACTOR="1"
python main.py --gui
```

User Note

After changing the environment variable, relaunch the graphical application and check whether the main workstation layout is readable and correctly fitted to the display.

9.2 Issue 2 — Neural Detector Falls Back to CPU

CUDA / CPU FALLBACK

Symptom

The runtime log reports that CUDA is unavailable and CPU inference is being used.

Likely Cause

A CUDA/PyTorch runtime mismatch or missing/incompatible NVIDIA drivers.

Documented Resolution

Install a PyTorch build compatible with the installed CUDA runtime and driver stack.

9.2.1 Supplied CUDA Installation Example

The command supplied in the original manual is:

```
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121
```

The command above is the installation example supplied with the HORIZON manual. It is not a universal command for every NVIDIA driver, CUDA runtime, or operating-system configuration. Verify the appropriate PyTorch/CUDA combination for the actual evaluation machine before changing the environment.

9.3 Issue 3 — High CPU Utilization During Monte Carlo Runs

BENCHMARK CPU LOAD

Symptom

UI responsiveness decreases while large parallel benchmark runs are executing.

Documented Cause

Worker-thread/process oversubscription.

Documented Resolution

Reduce the worker allocation through the `-threads` command-line option.

9.4 Operator FAQ

The following questions are derived from the functions already documented in Sections 1–8. They are intended as quick operational answers rather than new technical material.

9.4.1 How do I start the graphical application?

Use:

```
python main.py --gui
```

The application opens on the Mission Control screen.

9.4.2 Where should I begin a normal experiment?

Begin at **Mission Control**. Review the selected scenario, configuration, Experiment Preview, and Resolved Summary before starting execution.

The supplied user manual describes Mission Control as the pre-run workspace for selecting the experiment, resolving configuration, and previewing target motion. [:contentReference\[oaicite:5\]index=5](#)

9.4.3 Which workstation should I use while the experiment is running?

Use **Live Tracking** for real-time observation of the sensor view, PAT state, target position, and telemetry.

9.4.4 Where should I inspect unexpected tracking behavior?

Use **Track Diagnostics**. The documented workstation provides tracking-error history, covariance/innovation information, model probability information, and related diagnostic views.

9.4.5 Where do I apply disturbance conditions?

Use the **Disturbance Experiment Room**. The supplied controls include platform jitter, atmospheric turbulence, thermal noise, and impulse-noise settings.

9.4.6 Where do I run benchmark evaluations?

Use the **Scientific Benchmarking Workstation**, or use the documented headless `-benchmark` CLI mode.

9.4.7 How do I preserve a reproducible run record?

Retain the configuration, master seed, software revision, command used when applicable, and the generated output artifacts together. This is also the documented reproducibility guidance from the configuration and CLI sections.

9.4.8 Can I use an external MP4/AVI input?

Yes. HORIZON's documented external-video path supports MP4/AVI ingestion for the coarse-pointing evaluation workflow. The precise final GUI control label should be checked against the released executable because the supplied manual source does not define a button-by-button external-video sequence.

9.5 Known System Limitations & Future Scope

The following limitations and future extensions are taken directly from the supplied Part VII User Manual source. :contentReference[oaicite:6]index=6

Table 6: Known HORIZON system limitations and documented future extensions.

Area	Current Limitation	Documented Future Scope
FPA resolution	Fixed 256×256 pixel focal-plane representation for the current 100 Hz simulation profile.	GPU-accelerated support for larger sensor resolutions such as 512×512 and 1024×1024 .
Optical channel	Single monochromatic 850 nm optical channel.	Multi-spectral NIR/SWIR optical-path modelling.
Gimbal topology	Two-axis Pan-Tilt (2-DOF) rotation.	Three-axis gimbal with active line-of-sight roll stabilization.

User Note

A documented limitation defines the boundary of the current release. It should be treated as part of the operational record rather than silently assumed to be supported functionality.

9.6 Final Operator Check

Before closing an evaluation or laboratory session, verify:

SESSION CLOSE-OUT

1. Required run outputs have been retained.
2. The configuration associated with the run has been retained.
3. The master seed has been recorded where applicable.
4. Any troubleshooting action has been documented.
5. Known limitations encountered during the session have been recorded rather than omitted.